

● MEDIUM

● INFORMATION AND IDEAS

● ANSWER KEY

Information and Ideas

09

10 answers · Rationale-rich · Teach from doc

Q1**A**

- A. the current makeup of the Arabia Terra region might not accurately reflect the volcanic activity of Mars's past.
-

Choice A is the best answer because it presents the conclusion that most logically follows from the text's discussion of the Arabia Terra region of Mars. According to the text, there are craters in Arabia Terra that could be the result of volcanic activity, and researchers have found evidence of ash deposits consistent with a large eruption. The text goes on to note, however, that erosion and other events could have altered the surface of Mars. This observation suggests that current conditions on Mars's surface are not necessarily a reliable guide to past events—some signs of past events could have been transformed or erased entirely—and thus the current makeup of Arabia Terra may not accurately reflect past volcanic activity.

Choice B is incorrect because the text suggests only that past events could have changed Mars's surface such that its current appearance isn't a reliable guide to past activity, not that it's likely that past eruptions were not as massive as astronomers previously believed. Nothing in the text supports a conclusion about the likely size of past eruptions. Choice C is incorrect because the observation that resurfacing events could have changed the appearance of Mars doesn't imply that the ash discussed in the text likely came from multiple volcanoes. Although it's possible that the ash came from different volcanoes, there's no information in the text supporting a conclusion about how likely that possibility is. Choice D is incorrect because nothing in the text suggests that the Arabia Terra craters had to have been created by something other than volcanic eruptions. Although the text does suggest that the evidence consistent with volcanic eruptions shouldn't be taken as definitive proof of past eruptions, that doesn't mean that the craters couldn't have been created by eruptions, only that we can't be certain they were.

Q2

C

- C. CNO-treated females showed more activity relative to saline-treated females than CNO-treated males showed relative to saline-treated males.
-

Choice C is the best answer. The graph shows that the CNO-treated females were way more active than the CNO-treated males, while the saline-treated males and females (the control groups) had very similar activity levels. This supports the claim that there were sex-specific differences in the mice's response to neuron activation.

Choice A is incorrect. This choice misreads the graph. All four groups of mice started at nearly the same activity level before treatment (see how all four points are very close together at -4 minutes, meaning four minutes before treatment). Choice B is incorrect. This choice doesn't complete the assertion. The assertion is about the mice's response to neuron activation, so we need to include the data about the CNO-treated females and males. Choice D is incorrect. This choice misreads the graph. The line for the CNO-treated males does not show a "substantial decline" until around 122 minutes, and there is no "steep increase" afterward.

Q3

D

- D. Medina-Alcaide and her team tested Paleolithic light sources and learned some details about how Paleolithic artists traveled within dark caves.
-

Choice D is the best answer because it most accurately states the main idea of the text. The text indicates that archaeologist M^a Ángeles Medina-Alcaide and her team used replicas of Paleolithic light sources to understand how Paleolithic artists moved through dark caves. The researchers learned, for example, that torches were more helpful for moving through caves than animal-fat lamps were. Thus, the main idea of the text is that Medina-Alcaide and her team tested Paleolithic light sources and learned some details about how Paleolithic artists traveled within dark caves.

Choice A is incorrect because the text doesn't address the usefulness of fireplaces for Paleolithic cave artists; it only mentions fireplaces briefly as one of the three lighting methods the researchers tested. Choice B is incorrect because the text doesn't discuss how often Paleolithic cave artists used each kind of light source tested. Although the text does compare animal-fat lamps with another lighting method, the point of that comparison is that animal-fat lamps were less useful than torches when walking, not that one method was used more often than another. Choice C is incorrect because the text doesn't discuss either how difficult it was for Medina-Alcaide and her team to replicate light sources or how the team felt about drawing conclusions from their study. Instead, the text reveals that the team was able to conclude that each light source likely had a distinct purpose.

Q4

D

- D. illustrate both the relatively low scholarly regard in which ephemera was once held and the later recognition of ephemera's possible utility.
-

Choice D is the best answer. Johnson collected “ephemera,” or things that are meant to be thrown away. Scholars thought his collection was worthless to them, then later realized that it was potentially valuable. This suggests that scholars went from disregarding ephemera to recognizing their usefulness.

Choice A is incorrect. This inference isn't supported. The text tells us that the Bodleian Library acquired Johnson's large collection of ephemera back in 1968, so we can assume that contemporary historians conducting research there do have access to that collection. Choice B is incorrect. This inference isn't supported. The text tells us that “Oxford University's Bodleian Library acquired the collection,” but it never suggests that it was a challenge to do so. Choice C is incorrect. This inference isn't supported. The text actually suggests the opposite: the example of Johnson's collection lends support to arguments that ephemera does hold value for scholars.

Q5

B

- B. Vietnamese conveys information at approximately the same rate as Spanish despite being spoken at a slower rate.
-

Choice B is the best answer because it provides the most direct support from the table for the claim that two languages can convey similar amounts of information even if they're spoken at different rates. The table shows the approximate rates at which five languages are spoken and the rates at which those five languages convey information. Vietnamese is spoken at around 5.3 syllables per second, whereas Spanish is spoken at around 7.7 syllables per second, but the two languages convey information at very similar rates: Vietnamese at a rate of around 42.5 bits per second and Spanish at a rate of around 42.0 bits per second. Thus, the description of Vietnamese conveying information at around the same rate that Spanish does despite being spoken more slowly supports the claim in the text that languages can convey the same amount of information even if spoken at different rates.

Choice A is incorrect because it isn't true that Thai and Hungarian have the lowest rates of speech of the five languages shown. According to the table, Hungarian is spoken at around 5.9 syllables per second, which is faster than Vietnamese (5.3 syllables per second). Additionally, even if this statement were true, the assertion that two languages are spoken the slowest and convey information the slowest wouldn't support the claim that languages can convey the same amount of information even if they're spoken at different rates. Choice C is incorrect because it isn't true that the fastest-spoken language (Spanish, at 7.7 syllables per second) also conveys information the fastest: Spanish conveys information at 42.0 bits per second, which is slower than the 42.5 bits-per-second rate at which Vietnamese conveys information. Additionally, even if this statement were true, the assertion that the language spoken the fastest also conveys information the fastest has no bearing on the claim that languages can convey the same amount of information even if they're spoken at different rates. Choice D is incorrect because it isn't true that Serbian conveys information faster than Spanish does. According to the table, Serbian conveys information at a rate of around 39.1 bits per second, which is slower than the 42.0 bits-per-second rate at which Spanish conveys information.

Q6

C

- C. With the help of a special tool, researchers have determined the likely origin of bowls recovered from a shipwreck.
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Choice C is the best answer because it most accurately states the main idea of the text. According to the text, thousands of ceramic bowls were found in a recovered Chinese shipwreck. The text goes on to say that Niziolek and her team used a special tool, a portable X-ray fluorescence analyzer, to determine the bowls' chemical signatures. Comparing these chemical signatures with the chemical signatures of materials they had collected from old Chinese kiln sites, the text says, allowed the researchers to identify which kilns had produced the bowls. In other words, the researchers determined the bowls' origin.

Choice A is incorrect. Although the text indicates that the researchers used technology in the form of a portable X-ray fluorescence analyzer, it doesn't specifically state that this technology is new. In addition, the text says that Niziolek and her team used the tool to determine the chemical composition of bowls that were found in a Chinese shipwreck, not to locate and recover the shipwreck itself. There's no indication in the text that a new technology can help researchers locate and recover shipwrecks. Choice B is incorrect because the text indicates that the researchers collected materials from old kiln sites for chemical comparison with the ceramic bowls, which means that the researchers must have already known the location of those kiln sites. Rather than identifying the location of the kilns, the researchers determined which kilns in operation 1,000 years ago had likely produced the bowls that were found in the shipwreck.

Choice D is incorrect. Although the text says that using a portable X-ray fluorescence analyzer tool enabled Niziolek and her team to analyze artifacts in the form of ceramic bowls without damaging them, the text doesn't discuss how researchers analyzed artifacts before this tool was invented. Moreover, the point that the bowls were left undamaged isn't the text's main idea. Rather, it's a detail that's provided to develop the main idea, which is that the researchers used a special tool to determine where the bowls had been produced.

Q7

C

- C. *F. hypoleuca* nestlings' preference for their own dialect likely drives them when they mature to reproduce with other *F. hypoleuca* from local rather than nonlocal populations.
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Choice C is the best answer. Because “song preference plays a role in songbird mate selection,” and because *F. hypoleuca* nestlings display a preference for local dialects, we can infer that they will be more likely to choose mates from local populations.

Choice A is incorrect. We don't have any information suggesting that this preference disappears, so there's no basis for this inference. Choice B is incorrect. Although the passage discusses the number of begging calls made in response to various *F. hypoleuca* dialects, no mention is made about the amount of food received based on dialect preference. Therefore, there's no basis for this inference. Choice D is incorrect. There's no mention in the passage of methods of other types of local songbirds, so there's no basis for this inference.

Q8

A

- A. “Your true soul and body appear before me.”
-

Choice A is the best answer because it most directly illustrates the text's claim about Whitman's poem, “To You.” The text says that in this poem, Whitman suggests that he deeply understands the poem's reader. This quotation says that the reader's “true soul and body appear before” Whitman, thereby asserting that he can see the reader as the reader truly is, suggesting that he deeply understands the reader.

Choice B is incorrect because this quotation describes Whitman making the reader the subject of the poem (“you be my poem”), not Whitman deeply understanding the reader. Choice C is incorrect because instead of suggesting that Whitman deeply understands the reader, it emphasizes Whitman's regret at not having addressed the reader sooner. Choice D is incorrect. Although this quotation shows Whitman directly addressing the reader and expressing concern about the reader, it doesn't illustrate the idea that Whitman suggests that he deeply understands the reader. The quotation is simply expressing concern about the reader, which doesn't necessarily imply deep understanding of the reader.

Q9

A

A. He draws from traditional Native art forms to create his original works.

Choice A is the best answer because it most accurately describes Gibson's approach to art. As the text explains, Gibson, who is Cherokee and Choctaw, transforms punching bags into art pieces by applying (or attaching) to them beadwork and elements of Native dressmaking, including leather fringe and the jingles of the jingle dress. The text goes on to say that in most Native communities, the art forms of beadwork and dressmaking are traditionally practiced by women. Therefore, Gibson's approach to art consists of creating original works by drawing from traditional Native art forms.

Choice B is incorrect. Because Gibson incorporates Native art forms into his own original artwork, it can be inferred that he has been influenced by other Native artists, but the text never suggests that non-Native artists have influenced him. Choice C is incorrect because the text doesn't indicate that Gibson designs dresses influenced by boxing but instead that he turns punching bags, which are used in boxing, into works of art by applying elements of Native dressmaking to them. Choice D is incorrect. Although Gibson does incorporate beadwork into his art, the text never mentions the colors or patterns that he uses or suggests that his art defies the expectations that people might have about color and pattern in beadwork.

Q10

C

- C. depictions of food in Dutch paintings of the period should be taken as realistic representations of Dutch eating habits.
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Choice C is the best answer. The text tells us that lemons—an expensive imported product—feature in Dutch paintings of the period more frequently than dairy products do. Since it's unlikely lemons were eaten more often than dairy, this fact casts doubt on the theory that these paintings realistically depicted Dutch diets.

Choice A is incorrect. In fact, the text suggests the opposite: it says that dairy products were a "major component of the Dutch diet," while lemons were an expensive import item, so we can infer that lemons were eaten much more rarely than dairy products. Choice B is incorrect. The passage doesn't mention painters from other countries, so there's no basis to make this inference. Choice D is incorrect. In fact, the text suggests that Dutch painters do have reasons for depicting foods other than to show what Dutch people typically ate. For example, lemons in a painting could indicate wealth or prosperity.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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